

Dataset Understanding and Initial Analysis – Titanic Dataset

Objective:

The objective of this task is to understand the structure of the dataset, identify different types of features, check data quality, and evaluate whether the dataset is suitable for machine learning.

Dataset Description:

The Titanic dataset contains information about passengers such as age, gender, passenger class, fare, and survival status. It is commonly used for classification problems in machine learning.

Types of Data Identified:

- Numerical: Age, Fare, SibSp, Parch
- Categorical: Sex, Embarked, Ticket, Cabin
- Ordinal: Pclass
- Binary: Survived

Target Variable:

The target variable is **Survived**, which indicates whether a passenger survived (1) or not (0).

Data Quality Observations:

- The Age column contains missing values.
- The Cabin column has a large number of missing values.
- The dataset shows some imbalance between survived and non-survived passengers.

Conclusion:

The dataset is suitable for machine learning, but it requires data cleaning and preprocessing, especially handling missing values and encoding categorical variables, before building any predictive model.